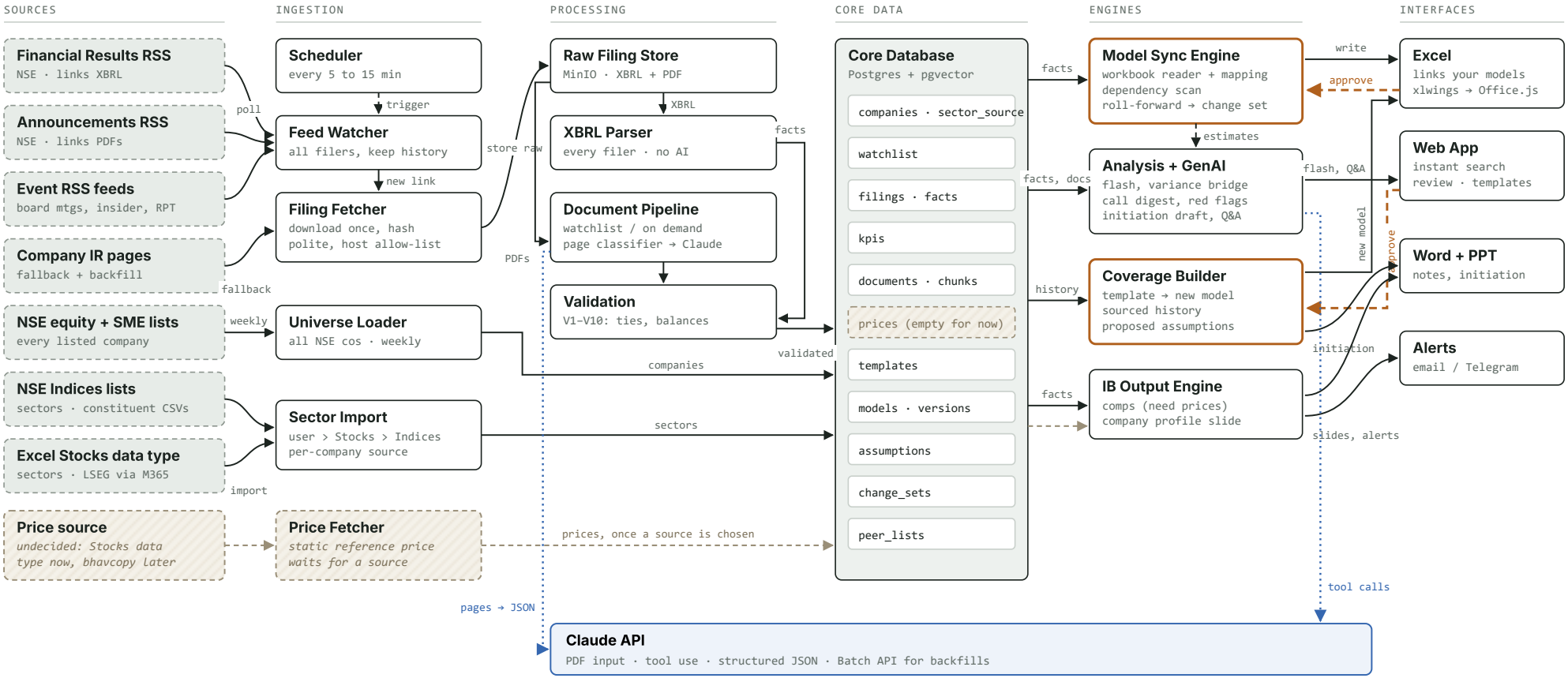
Analekh Architecture

The analyst's record, kept for you.

Every NSE-listed company is preloaded and its results are parsed as they are filed. For a company you follow, the system finds and updates your Excel model after you approve each change, or builds a new model from your sector template and drafts an initiation. Core numbers come from XBRL; the AI writes the analysis and pulls every figure from the database. The price source is still undecided.

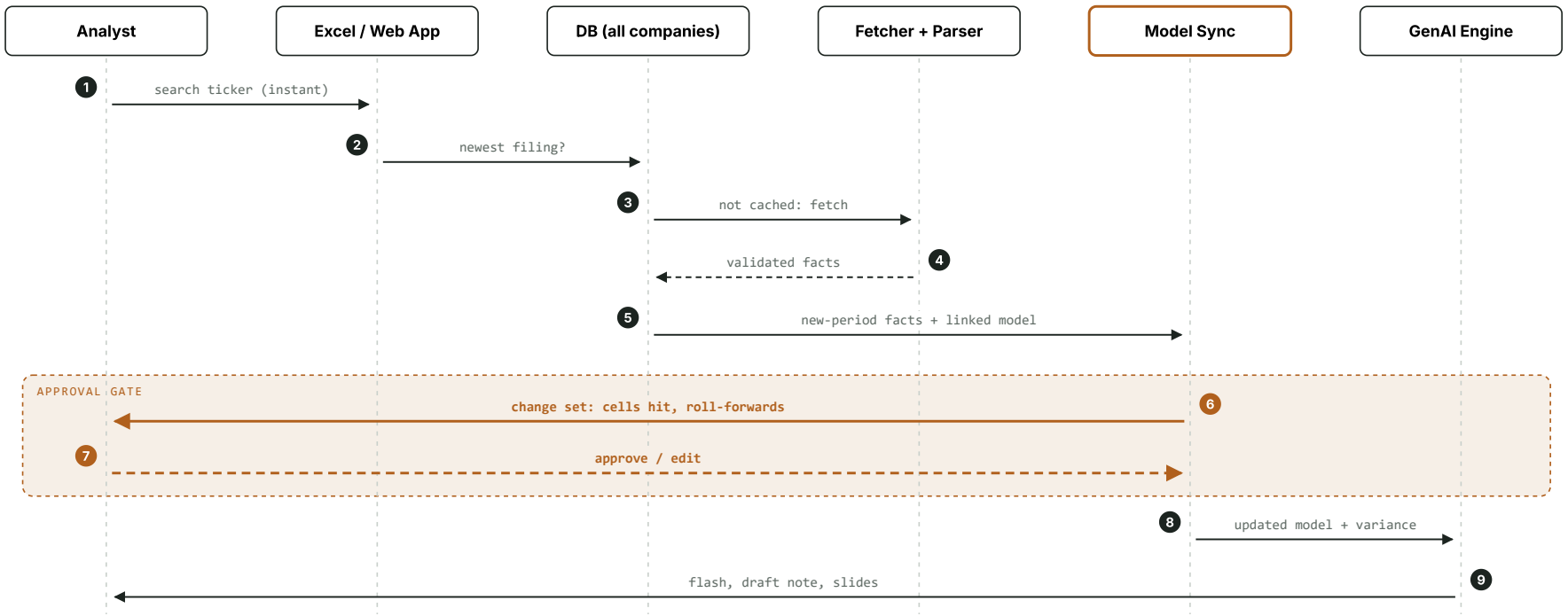
Components and data flow



data · control / return · analyst approval · Claude API call · pending (no source yet) · external source · undecided source

The NSE equity and SME lists preload every company and NSE's RSS feeds are polled on a schedule, so results for all filers are parsed automatically and "newest filing for this ticker" is a database lookup. Sectors are imported by precedence (user override, then Excel's Stocks data type, then NSE Indices lists) and recorded per company. Only the relevant PDF pages reach Claude. Nothing is written to Excel until the analyst approves.

What happens when an analyst enters a ticker

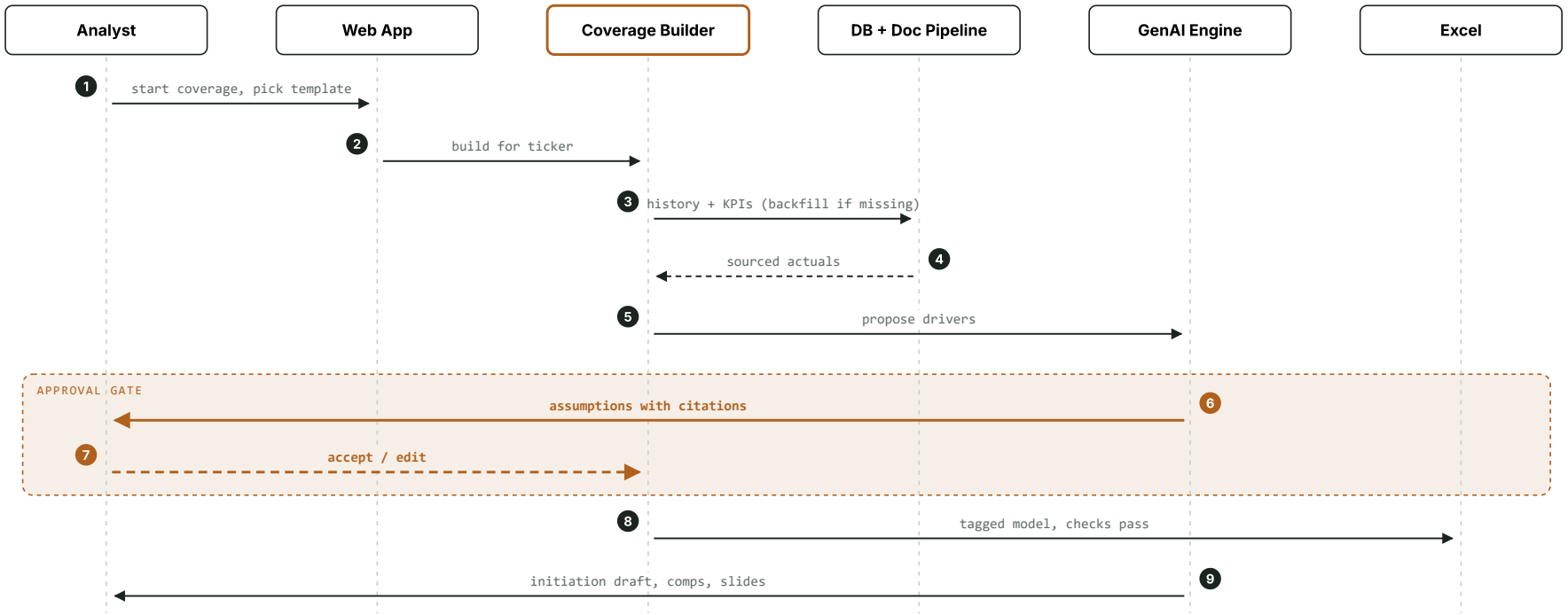


The ticker search runs in the browser over the whole universe, so the company opens at once. Steps 3 and 4 run only when the filing isn't already in the database. The linked model is found through watched folders or the open workbook, identified by a hidden tag, its file name or its contents. Steps 6 and 7 are the approval gate: the analyst sees which cells change and which forecasts need rolling forward, accepts or edits each one, and only then is a new model version saved.

What changed in v0.0.2

- **Prices:** Kite Connect removed. Undecided: Excel's Stocks data type in the workbook now, NSE's end-of-day bhavcopy later; one static reference price at a time.
- **Sectors:** user override, then the Stocks data type (LSEG industry), then NSE Indices constituent lists; the source is recorded per company.
- **Search:** ranked type-ahead; the universe loads once and is searched in the browser.
- **Build and CI:** GitLab Free with OSV-Scanner; pnpm workspace, Node 24.
- **Versions** follow the build phases: v0.0.x in Phase 0, v0.1.x once it is complete; v1.0.0-pre-release is the first release for testing.

What happens when an analyst starts coverage



Used when no model exists yet. The analyst picks a company and a sector template (their own upload or a built-in one). The builder fills the template's inputs with sourced history and keeps its formulas. Proposed assumptions pass through the same approval gate before the tagged model is saved. From the next results onward, the quarterly update flow takes over.

Numbers come from data

Figures in drafts are fetched from the database or the model through tool calls, and each one carries its XBRL tag or PDF page reference.

Data rights

NSE's published RSS feeds at their stated interval, its published lists, NSE Indices' constituent CSVs and company IR pages. No NSE internal endpoints. Stocks data type fields are for your own use. Written confirmation or a licence before any public release.

Your templates

Upload a model template per sector. The system reads its structure, you confirm the mapping once, and a test build must pass before it is used.

Prices (undecided)

One static reference price at a time: the day's close after the market closes, the day's open during market hours. Stocks data type in Excel now; NSE's bhavcopy, which is end of day only, later.

Runs from India

NSE's main site is geo-restricted, so the scheduler and fetchers run on an Indian network or VM. When NSE hosts stop answering, fetchers retry later with single requests.

Stack

`Python 3.12 · FastAPI · Celery + Redis · Arelle · PyMuPDF · Claude API · Postgres + pgvector · MinIO · xlwings · React + pnpm · Docker Compose · GitLab CI`